

24, Oehang 1-gil, Gunsan-si, Jeollabuk-do, Republic of Korea		Rev. date	2022. 10. 04
Tel. 82- 63- 462- 6018	HK-940	Version	5
Fax 82- 63- 462- 6009	CAS No : 9003-01-4	page	5

1. Chemical product and company identificaton

- A. Product Name **HK-940**
- B. INCI Name Carbomer
- C. Material Uses Thickener
- D. Supplier/Manufacturer

HANNONG CHEMICALS

24, Oehang 1-gil, Gunsan-si, Jeollabuk-do, Republic of Korea

82- 63- 462- 6018

R&D center

2. Hazards Identification

- A. GHS classification of the substance
- No particular hazards known.

- B. GHS label elements, including precautionary statements

☐ Symbols	☐ Signal word	☐ Hazard statement
		☐ Precautionary statements Avoid contactwith skin, eyes or clothing. Wash eyes immediately with large amounts of water and get medical attention immediately Wear suitable clothes Wear glasses/masks.

- C. Other hazards which do not result in classification or are not covered by the GHS

-NFPA Health : 1 Fire Hazard : 0 Reactivity : 0

3. Composition/information on ingredients

A. Substances

Chemical Name	CAS No.	Contents(w/w%)
Poly(Acrylic acid)	9003-01-4	>99.9%
Benzene	71-43-2	<0.1%

4. First aid measures

A. General advice:

Remove contaminated clothing.

B. If inhaled:

Remove the affected individual into fresh air and keep the person calm. Seek medical attention.

C. If on skin:

Wash thoroughly with soap and water.

D. If in eyes:

Seek medical attention.

Wash affected eyes for at least 15 minutes under running water with eyelids held open.

E. If in eyes:

Seek medical attention.

Wash affected eyes for at least 15 minutes under running water with eyelids held open.

F. If swallowed:

Immediately rinse mouth and then drink plenty of water, do not induce vomiting. Seek medical attention.

5. Fire-fighting measures

A. Suitable extinguishing media:

CO₂, foam, dry extinguishing media, water spray

B. Products of Combustion : Carbon oxide(CO, CO₂)

C. Protective equipment for fire-fighting:

Cool containers with water spray until well after the fire is out

Fire may be spread by water use

Do not scatter spilled material with high-pressure water streams

6. Accidental release measures

A. Personal precautions:

Avoid contacting with skin, eyes, clothing.

Keep away from sources of ignition

Do not breathe dust. Wear the dust-protection mask. Prevent dust accumulation

Prevent dust accumulation

B. Environmental precautions:

Do not allow materials to enter sewers, soil or surface water.

C. Methods for cleaning up :

- For small spills

Vacuum up spillage and collect in suitable containers for disposal.

- For large spills

Collect dust using a vacuum cleaner equipped with HEPA filter. Prevent entry into waterways, sewers, basements or confined areas. Following product recovery, flush area with water

7. Handling and storage

A. Handling

Avoid contact with eyes. Do not ingest. Avoid breathing vapors or spray mists.

Avoid prolonged or repeated contact with skin. Do not get on skin or clothing.

Keep container closed. Wash thoroughly after handling.

B. Storage

Keep away from sources of ignition.

Container must be tightly closed and stored out of contact with water.

8. Exposure controls / personal protection

A. Control parameters.

Occupational exposure or biological limit values

None of the components have assigned exposure limits

Other exposure limits

Chemical Name	Type	Exposure Limit Values	Source
Polyacrylic acid	TWA	3mg/m ³	

B. Engineering controls:

Sufficient ventilation to minimize vapor exposure.

C. Personal protection:

Eyes : Safety glasses with side-shields.

Body : Lab coat.

Hands : Gloves.

Respiratory : Wear appropriate respirator when ventilation is inadequate.

9. Physical and chemical properties

A. Appearance	Solid, White powder	B. Odour	Slightly Acetic
C. Odour threshold	Not available	D. pH (1%Aqueous solution)	2.5 - 3.5
E. mp/fp(°C)	Not available	F. bp(°C)	Not available
G. Flash point(°C, open cup)	Not applicable	H. Evaporation rate	min. 130g/m ³
I. Flammability(solid,gas)	Non-Flammable Solid	J. Upper/lower explosive limits	Not available
K. Vapor pressure	Not available	L. Solubility in water	swellable
M. Vapor density(air=1)	Not available	N. Relative density(@25°C)	1.4
O. Partition coefficient(n-octanol/water)	Not available	P. Autoignition temp.(°C)	Approx. 520
Q. Decomposition temp.(°C)	Not available	R. Viscosity(0.2% in water)	< 10,000cps
S. Bulk density(g/ml, @25°C)	< 0.24	T. Percent volatile(%)	< 2 by weight
U. Molecular Weight	> Mw 3,000,000		

10. Stability and reactivity

- A. Chemical stability : Stable under normal storage conditions.
- B. Incompatibility with various substances : oxidizing agents. Strong alkalies.
- C. Possibility of hazardous reactions : Material is stable
- D. Hazardous decomposition products : None when used as directed.

Thermal decomposition will lead to the release of CO & CO₂

11. Toxicological information

- A. Information on the likely routes of exposure

- inhalation : No data available
- ingestion : Nausea, vomiting, diarrhea, headache, drowsiness, dizziness
- eye : No data available
- skin : No data available

- B. Delayed and immediate effects and also chronic effects from short- and long-term exposure

Acute toxicity	oral	LD50 2500 mg/kg Rat
Skin irritation	Not irritating	
Eye irritation	Not irritating	
Chronic toxicity	Not available	

- C. Numerical measures of toxicity(such as acute toxicity estimates) : Not available

12. Ecological information

A. Ecotoxicity

Fish

Benzene : LC50 - Oncorhynchus mykiss - 5.3 mg/l - 96 h

Aquatic Invertebrates

Benzene : EC 50 (Water Flea (Daphnia magna), 2 d): 10mg/L

Toxicity to Aquatic Plants

Benzene : EC 50 (Green algae (Selenastrum capricornutum), 3 d): 100 mg/L

B. Persistence and degradability

Biodegradation

Benzene : IC50 - 13mg/l - 24h

C. Bioaccumulative potential

Bioconcentration Factor (BCF) : Benzene - 10

Partition Coefficient n-octanol / water (log Kow)

Benzene : Log Kow: 2.24

Mobility : No data available

D. Other adverse effects : No data available

13. Disposal considerations

A. Waste information

Waste must be disposed of in accordance with federal, state and local environmental control regulations.

B. Container disposal

Dispose of in a licensed facility. Recommend crushing, puncturing or other means to prevent unauthorized use of used containers.

14. Transport information

A. UN number	:	Not classified
B. UN proper shipping name	:	Not classified
C. Transport Hazsrd class(es)	:	Not classified
D. Packing group	:	Not classified
E. Marine pollutant	:	No
F. ICAO/IATA	:	NOT REGULATED

15. Regulatory information

A. U.S. Regulation (TSCA)	:	listed
B. Other International lists	:	Korea(TCCL) listed

16. Other information

A. Information on prepatation

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B. Revision date & version 2022. 10. 04